Kirinyaga University
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SCHOOL: School of Pure and Applied Sciences

DEPARTMENT: **Pure and Applied Sciences**

PROGRAMME: Bachelor of Science in Software Engineering

YEAR: 3 **SEMESTER:** 1

UNIT CODE: SSE 2105 **UNIT TITLE:** Advanced Object-Oriented Programming

LECTURE HOURS: 45 **PRE-REQUISITES:** None

LECTURER'S NAME: Mr. David Gichohi Maina

LECTURER'S CONTACTS; TEL: 0708398130 **EMAIL:** dmaina@kyu.ac.ke

Welcome Message

Welcome to the course SSE 2105: Advanced Object Oriented Programming that enables students use object-oriented design methods and programming techniques to produce solutions for a wide range of problem specifications.

Learning Outcomes

1. To learn the basic syntax and semantics of the Java language.
2. To understand the concepts of classes and objects.
3. Design, code, and implement, simple to intermediate level standalone OOP applications

Teaching Methodology

- Blended learning - combines both online and physical classroom learning
- Case analysis
- Group discussions

Instructional Materials

- White board
- LCD Projector
- Smarboards
- Masomo Portal

Course Evaluation

- CAT 1 = 15 %
- CAT 2 = 15 %
- Assignments = 10 %
- Examination = 70%
- Pass mark = 40 %

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COURSE OUTLINE

S/N.	Week	Sub-Topic	Remarks
1.	Introduction to object-oriented programming paradigm	• Definitions- encapsulation, abstraction, classes, objects, polymorphism, inheritance, data hiding. • Examples of object-oriented programming languages. • Merits and demerits of OOP. • Comparison between procedure-oriented programming and OOP • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
2.	Introduction to Java Programming	• Why learn Java, application & benefits. • Overview of Java, installing & setting up Java • Java basic concepts; identifiers & keywords, Java data types, operators, expressions, namespace, type casting, I/O statements. • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
3.	Java Control Structures	• Decision structures- if, if..else, nested if. • Loops – for, while, do..while, nested loops. • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
4.	Java Functions & Arrays	• Declaring and defining functions • Calling functions • Parameters and parameter passing • Declaring and defining arrays- One dimensional and multidimensional arrays • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
5.	Week 5	• CAT ONE/ ASSIGNMENT ONE	
6.	Objects and Classes	• Introduction • Defining class in Java • Data members, member functions, access specifiers, • Creating objects in Java.	

		• Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
7.	Java Inheritance	• Introduction • Types of inheritance • Implementing inheritance in Java • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
8.	Java Polymorphism	• Introduction • Method and operator overriding and overloading • Implementing polymorphism in Java • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
9.	Java Encapsulation and Abstraction	• Introduction • Why encapsulation? • Ways of achieving abstraction • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
10.	Java Input/Output	• Creating and using GUIs, • GUI Components, • Adding components to containers • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
11.	Week 10	• **CAT TWO/ ASSIGNMENT TWO**	
12.	Java Applets	• Introduction to Java Applets, • Drawing strings, applets, lines and other basic shapes, • Putting applets on the web. • Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
13.	Advanced Java Programming	• Java database connection, Data insertion, retrieval, updates and deletion from Java forms to database.	

		• Topic Activity (Group Discussions, summarize every week's topic on what you have learnt & practice exercises)	
14.	Week 13	• **REVISION FOR EXAMS**	
15.	Week 14	• **END OF SEMESTER EXAMINATIONS**	

COURSE TEXTBOOKS

1. Savitch W.(2011). Java an Introduction to Problem Solving and Programming, Prentice Hall. ISBN : 978- 0534359492.
2. Savitch, W. (2009). Java – An introduction to problem solving and programming (6th Ed), Prentice Hall ISBN : 978- 046783222492.
3. Deitel and Deitel. (2010) Java. How to Program by (6th Ed). ISBN: 978- 6789023456.

REFERENCE TEXTBOOKS

1. Savitch W. (2011). Java an Introduction to Problem Solving and Programming, Prentice Hall. ISBN : 978- 078909492.
2. Barry J. Holmes, Daniel T. Joyce. (2000). Object Oriented Programming with Java. Java Software Solutions- Foundations of Program Design- John Lewis, William Loftus, ISBN : 978- 35678959492.

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